is defined as

$$A = 10\log\left[\frac{S_{av}(z)}{S_{av}(0)}\right] = 10\log(e^{-2\alpha z}) = -20\alpha z\log e = -8.68\alpha z = -\alpha\ [\text{dB/m}]\ z \quad \text{(dB)}, \tag{7.113}$$

where

$$\alpha\ [\text{dB/m}] = 8.68\alpha \quad \text{(Np/m)}. \tag{7.114}$$

Since $\mathbf{S}_{av}(z)$ is directly proportional to $|\mathbf{E}(z)|^2$,

$$A = 10\log\left[\frac{|E(z)|^2}{|E(0)|^2}\right] = 20\log\left[\frac{|E(z)|}{|E(0)|}\right] \quad \text{(dB)}. \tag{7.115}$$

Example 7-6: Power Received by a Submarine Antenna

A submarine at a depth of 200 m below the sea surface uses a wire antenna to receive signal transmissions at 1 kHz. Determine the power density incident upon the submarine antenna due to the EM wave of Example 7-4.

Solution: From Example 7-4, $|\tilde{E}(0)| = |E_{x0}| = 4.44$ (mV/m), $\alpha = 0.126$ (Np/m), and $\eta_c = 0.044\angle 45^\circ$ (Ω). Application of Eq. (7.109) gives

$$\mathbf{S}_{av}(z) = \hat{\mathbf{z}}\frac{|E_0|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta = \hat{\mathbf{z}}\frac{(4.44\times10^{-3})^2}{2\times0.044}e^{-0.252z}\cos 45^\circ = \hat{\mathbf{z}}0.16e^{-0.252z} \quad \text{(mW/m}^2\text{)}.$$

At $z = 200$ m, the incident power density is

$$\mathbf{S}_{av} = \hat{\mathbf{z}}(0.16\times10^{-3}e^{-0.252\times200}) = 2.1\times10^{-26} \quad \text{(W/m}^2\text{)}.$$

Exercise 7-10: Convert the following values of the power ratio G to decibels: (a) 2.3, (b) 4×10^3, and (c) 3×10^{-2}.

Answer: (a) 3.6 dB, (b) 36 dB, (c) −15.2 dB. (See EM.)

Exercise 7-11: Find the voltage ratio g corresponding to the following decibel values of the power ratio G: (a) 23 dB, (b) −14 dB, and (c) −3.6 dB.

Answer: (a) 14.13, (b) 0.2, (c) 0.66. (See EM.)

Chapter 7 Summary

Concepts

- A spherical wave radiated by a source becomes approximately a uniform plane wave at large distances from the source.
- The electric and magnetic fields of a transverse electromagnetic (TEM) wave are orthogonal to each other, and both are perpendicular to the direction of wave travel.
- The magnitudes of the electric and magnetic fields of a TEM wave are related by the intrinsic impedance of the medium.
- Wave polarization describes the shape of the locus of the tip of the **E** vector at a given point in space as a function of time. The polarization state, which may be linear, circular, or elliptical, is governed by the ratio of the magnitudes of and the difference in phase between the two orthogonal components of the electric field vector.
- Media are classified as lossless, low-loss, quasi-conducting, or good conducting on the basis of the ratio $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon$.
- Unlike the dc case, where the current flowing through a wire is distributed uniformly across its cross section, the ac case has most of the current is concentrated along the outer perimeter of the wire.
- Power density carried by a plane EM wave traveling in an unbounded medium is akin to the power carried by the voltage/current wave on a transmission line.

定义为

$$A = 10\log\left[\frac{S_{av}(z)}{S_{av}(0)}\right] = 10\log(e^{-2\alpha z}) = -20\alpha z\log e = -8.68\alpha z = -\alpha\ [\text{dB/m}]\ z \quad \text{(dB)}, \tag{7.113}$$

在哪里

$$\alpha\ [\text{dB/m}] = 8.68\alpha \quad \text{(Np/m)}. \tag{7.114}$$

由于 $S_{av}(z)$ 与 $|E(z)|^2$ 成正比，

$$A = 10\log\left[\frac{|E(z)|^2}{|E(0)|^2}\right] = 20\log\left[\frac{|E(z)|}{|E(0)|}\right] \quad \text{(dB)}. \tag{7.115}$$

例 7-6： 海底天线接收到的功率

位于海面以下 200 m 深度的潜艇使用线天线接收 1 kHz 的信号传输。确定例 7-4 中的电磁波作用在海底天线上的功率密度。

解：根据例 7-4，$|\tilde{E}(0)| = |E_{x0}| = 4.44$ (mV/m)、$\alpha = 0.126$ (Np/m) 和 $\eta_c = 0.044\angle 45^\circ$ (Ω)。方程的应用(7.109) 给出

$$\mathbf{S}_{av}(z) = \hat{\mathbf{z}}\frac{|E_0|^2}{2|\eta_c|}e^{-2\alpha z}\cos\theta_\eta = \hat{\mathbf{z}}\frac{(4.44\times10^{-3})^2}{2\times0.044}e^{-0.252z}\cos 45^\circ = \hat{\mathbf{z}}0.16e^{-0.252z} \quad \text{(mW/m}^2\text{)}.$$

在 z = 200 m 处，入射功率密度为

$$\mathbf{S}_{av} = \hat{\mathbf{z}}(0.16\times10^{-3}e^{-0.252\times200}) = 2.1\times10^{-26} \quad \text{(W/m}^2\text{)}.$$

练习7-10：将下列功率比G值转换为分贝：(a) 2.3、(b) 4×10^3和(c) 3×10^{-2}。

Answer: (a) 3.6 dB, (b) 36 dB, (c) −15.2 dB. (See EM.)

练习7-11：求与以下功率比G的分贝值对应的电压比g：(a)23dB，(b)−14dB，(c)−3.6dB。

答案：(a)14.13，(b)0.2，(c)0.66。 （参见EM。）

第7章总结

概念

- 由源辐射的球面波在距源较远的距离处近似变为均匀的平面波。
- 横向电磁（TEM）波的电场和磁场彼此正交，并且都垂直于波传播的方向。
- TEM 波的电场和磁场的大小与介质的固有阻抗相关。
- 波偏振描述了空间中给定点处 E 矢量尖端轨迹的形状随时间的变化。偏振态可以是线性、圆形或椭圆形，由以下比率决定：电场矢量的两个正交分量之间的幅度和相位差。
- 根据比率 $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon$，介质被分类为无损、低损耗、准传导或良好传导。
- 与直流情况不同，流经电线的电流在其横截面上均匀分布，而交流情况则大部分电流集中在电线的外周上。
- 在无界介质中传播的平面电磁波所承载的功率密度类似于传输线上的电压/电流波所承载的功率。

Mathematical and Physical Models

Complex Permittivity

$$\varepsilon_c = \varepsilon' - j\varepsilon''$$
$$\varepsilon' = \varepsilon$$
$$\varepsilon'' = \frac{\sigma}{\omega}$$

Lossless Medium

$$k = \omega\sqrt{\mu\varepsilon}$$
$$\eta = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega)$$
$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s})$$
$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})$$

Wave Polarization

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\,\hat{\mathbf{k}} \times \widetilde{\mathbf{E}}$$
$$\widetilde{\mathbf{E}} = -\eta\,\hat{\mathbf{k}} \times \widetilde{\mathbf{H}}$$

Maxwell's Equations for Time-Harmonic Fields

$$\nabla \cdot \widetilde{\mathbf{E}} = 0$$
$$\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}$$
$$\nabla \cdot \widetilde{\mathbf{H}} = 0$$
$$\nabla \times \widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}$$

Lossy Medium

$$\alpha = \omega\left\{\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}-1\right]\right\}^{1/2} \quad (\text{Np/m})$$
$$\beta = \omega\left\{\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}+1\right]\right\}^{1/2} \quad (\text{rad/m})$$
$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2} \quad (\Omega)$$
$$\delta_s = \frac{1}{\alpha} \quad (\text{m})$$

Power Density

$$\mathbf{S}_{av} = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^*\right] \quad (\text{W/m}^2)$$

Important Terms

Provide definitions or explain the meaning of the following terms:

attenuation constant α
attenuation rate A
auxiliary angle ψ_0
average power density $\mathbf{S}_{av}$
axial ratio
circular polarization
complex permittivity ε_c
dc and ac resistances
elliptical polarization
ellipticity angle χ
good conductor
guided wave
homogeneous wave equation
in phase
inclination angle
internal or surface impedance
intrinsic impedance η
LHC and RHC polarizations
linear polarization
lossy medium
low-loss dielectric
out of phase
phase constant β
phase velocity
polarization state
Poynting vector $\mathbf{S}$
propagation constant γ
quasi-conductor
rotation angle γ
skin depth δ_s
spherical wave
surface resistance R_s
TEM wave
unbounded
unbounded wave
uniform plane wave
wave polarization
wavefront
wavenumber k

332 CHAPTER 7 PLANE-WAVE PROPAGATION

数学与物理模型

复介电率

$$\varepsilon_c = \varepsilon' - j\varepsilon''$$
$$\varepsilon' = \varepsilon$$
$$\varepsilon'' = \frac{\sigma}{\omega}$$

无损介质

$$k = \omega\sqrt{\mu\varepsilon}$$
$$\eta = \sqrt{\frac{\mu}{\varepsilon}} \quad (\Omega)$$
$$u_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s})$$
$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})$$

波的极化

$$\widetilde{\mathbf{H}} = \frac{1}{\eta}\,\hat{\mathbf{k}} \times \widetilde{\mathbf{E}}$$
$$\widetilde{\mathbf{E}} = -\eta\,\hat{\mathbf{k}} \times \widetilde{\mathbf{H}}$$

时谐场的麦克斯韦方程组

$$\nabla \cdot \widetilde{\mathbf{E}} = 0$$
$$\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}$$
$$\nabla \cdot \widetilde{\mathbf{H}} = 0$$
$$\nabla \times \widetilde{\mathbf{H}} = j\omega\varepsilon_c\widetilde{\mathbf{E}}$$

有损介质

$$\alpha = \omega\left\{\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}-1\right]\right\}^{1/2} \quad (\text{Np/m})$$
$$\beta = \omega\left\{\frac{\mu\varepsilon'}{2}\left[\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}+1\right]\right\}^{1/2} \quad (\text{rad/m})$$
$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2} \quad (\Omega)$$
$$\delta_s = \frac{1}{\alpha} \quad (\text{m})$$

平均功率密度

$$\mathbf{S}_{av} = \tfrac{1}{2}\,\mathfrak{Re}\left[\widetilde{\mathbf{E}} \times \widetilde{\mathbf{H}}^*\right] \quad (\text{W/m}^2)$$

重要术语

请给出下列术语的定义或解释其含义。

衰减常数 α
衰减率 A
辅助角 ψ_0
平均功率密度 S_{av}
轴比
圆极化
复介电率 ε_c
直流电阻与交流电阻
椭圆极化
椭圆率角 χ
良导体
导行波
齐次波动方程
同相
倾角
内阻抗或表面阻抗
本征阻抗 η
左旋圆极化与右旋圆极化
线极化
有损介质
低损耗电介质
异相
相位常数 β
相速度
极化状态
坡印廷矢量 S
传播常数 γ
准导体
旋转角 γ
趋肤深度 δ_s
球面波
表面电阻 R_s
TEM波
无界
无界波
均匀平面波
波的极化
波前
波数 k

适用范围：上表采用 $e^{j\omega t}$ 时间约定；场方程用于无外加源的线性、均匀、各向同性介质，传导电流并入复介电率。电、磁场叉积关系表示平面波的场方向与波阻抗关系；有损介质须使用复波阻抗 η_c。

PROBLEMS

Section 7-2: Propagation in Lossless Media

7.1 The magnetic field of a wave propagating through a certain nonmagnetic material is given by

$$\mathbf{H} = \hat{\mathbf{z}}\,30\cos(10^8 t - 0.5y) \qquad \text{(mA/m)}.$$

Find:

*(a) the direction of wave propagation,

(b) the phase velocity,

*(c) the wavelength in the material,

(d) the relative permittivity of the material, and

(e) the electric field phasor.

7.2 Write general expressions for the electric and magnetic fields of a 1 GHz sinusoidal plane wave traveling in the $+y$ direction in a lossless nonmagnetic medium with relative permittivity $\varepsilon_r = 9$. The electric field is polarized along the x direction, its peak value is 6 V/m, and its intensity is 4 V/m at $t = 0$ and $y = 2$ cm.

7.3 The electric field phasor of a uniform plane wave is given by $\tilde{\mathbf{E}} = \hat{\mathbf{y}}\,10e^{j0.2z}$ (V/m). If the phase velocity of the wave is 1.5×10^8 m/s and the relative permeability of the medium is $\mu_r = 2.4$, find:

*(a) the wavelength,

(b) the frequency f of the wave,

(c) the relative permittivity of the medium, and

(d) the magnetic field $\mathbf{H}(z,t)$.

7.4 The electric field of a plane wave propagating in a nonmagnetic material is given by

$$\mathbf{E} = [\hat{\mathbf{y}}\,3\sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}}\,4\cos(\pi \times 10^7 t - 0.2\pi x)] \qquad \text{(V/m)}$$

Determine:

(a) the wavelength

(b) ε_r

(c) $\mathbf{H}$

*7.5 A wave radiated by a source in air is incident upon a soil surface, where a part of the wave is transmitted into the soil medium. If the wavelength of the wave is 60 cm in air and 20 cm in the soil medium, what is the soil's relative permittivity? Assume the soil to be a very low-loss medium.

*Answer(s) available in Appendix E.

7.6 The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\varepsilon_r = 2.56$ is given by

$$\mathbf{E} = \hat{\mathbf{y}}\,20\cos(6\pi \times 10^9 t - kz) \qquad \text{(V/m)}.$$

Determine:

(a) f, u_p, λ, k, and η.

(b) the magnetic field $\mathbf{H}$.

7.7 The magnetic field of a plane wave propagating in a nonmagnetic material is given by

$$\mathbf{H} = \hat{\mathbf{x}}\,60\cos(2\pi \times 10^7 t + 0.1\pi y) + \hat{\mathbf{z}}\,30\cos(2\pi \times 10^7 t + 0.1\pi y) \qquad \text{(mA/m)}.$$

Determine:

*(a) the wavelength

(b) ε_r

(c) $\mathbf{E}$

7.8 A 60 MHz plane wave traveling in the $-x$ direction in dry soil with relative permittivity $\varepsilon_r = 4$ has an electric field polarized along the z direction. Assuming dry soil to be approximately lossless, and given that the magnetic field has a peak value of 10 (mA/m) and that its value was measured to be 7 (mA/m) at $t = 0$ and $x = -0.75$ m, develop complete expressions for the wave's electric and magnetic fields.

Section 7-3: Wave Polarization

*7.9 An RHC-polarized wave with a modulus of 2 (V/m) is traveling in free space in the negative z direction. Write the expression for the wave's electric field vector given that the wavelength is 6 cm.

7.10 For a wave characterized by the electric field

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,a_x\cos(\omega t - kz) + \hat{\mathbf{y}}\,a_y\cos(\omega t - kz + \delta),$$

identify the polarization state, determine the polarization angles (γ, χ), sketch the locus of $\mathbf{E}(0,t)$, and verify your solution using Module 7.3 for each of the following cases.

(a) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 0$

(b) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 180°$

(c) $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = 45°$

(d) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -135°$

问题

第 7-2 节：无损媒体中的传播

7.1 通过某种非磁性材料传播的波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{z}}\,30\cos(10^8 t - 0.5y) \qquad \text{(mA/m)}.$$

寻找：

*(a) 波传播的方向，

(b) 相速度，

*(c) 材料中的波长，

(d) 材料的相对介电常数，以及 (e) 电场相量

。

7.2 写出 1 GHz 正弦平面波在相对介电常数为 $\varepsilon_r =$ 的无损非磁性介质中沿 +y 方向传播的电场和磁场的一般表达式。 9. 电场沿 x 方向极化，其峰值为 6 V/m，在 t = 0 和 y = 2 cm 处强度为 4 V/m。

7.3 均匀平面波的电场相量由 $\tilde{\mathbf{E}} = \hat{\mathbf{y}}\,10e^{j0.2z}$ (V/m) 给出。如果波的相速度为 1.5×10^8 m/s，介质的相对磁导率为 $\mu_r = 2.4$，则可得：

*(a) 波长，

(b) 波的频率 f，

(c) 介质的相对介电常数，以及

(d) 磁场 H(z,t)。

7.4 在非磁性材料中传播的平面波的电场由下式给出

$$\mathbf{E} = [\hat{\mathbf{y}}\,3\sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}}\,4\cos(\pi \times 10^7 t - 0.2\pi x)] \qquad \text{(V/m)}$$

决定：

(a) 波长 (b) ε_rc) H

(

*7.5 空气中的源辐射的波入射到土壤表面，一部分波传输到土壤介质中。如果波的波长在空气中为 60 厘米，在土壤介质中为 20 厘米，那么土壤的相对介电常数是多少？假设土壤是损耗非常低的介质。

*Answer(s) available in Appendix E.

7.6 在 $\varepsilon_r = 2.56$ 的无损、非磁性介电材料中传播的平面波的电场由下式给出

$$\mathbf{E} = \hat{\mathbf{y}}\,20\cos(6\pi \times 10^9 t - kz) \qquad \text{(V/m)}。$$

决定：

(a) f 、u_p、λ 、 k 和 η

。 (b) 磁场H。

7.7 在非磁性材料中传播的平面波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{x}}\,60\cos(2\pi \times 10^7 t + 0.1\pi y) + \hat{\mathbf{z}}\,30\cos(2\pi \times 10^7 t + 0.1\pi y) \qquad \text{(mA/m)}.$$

决定：

*(a) 波长 (b) ε_rc) E

(

7.8 60 MHz 平面波在相对介电常数为 $\varepsilon_r = 4$ 的干燥土壤中沿 −x 方向传播，具有沿 z 方向极化的电场。假设干燥土壤大致无损耗，并且磁场的峰值为 10 (mA/m)，并且在 t = 0 和 x = −0.75 m 处测得其值为 7 (mA/m)，请建立波电场和磁场的完整表达式。

第 7-3 节：波偏振

*7.9 模数为 2 (V/m) 的 RHC 偏振波在自由空间中沿负 z 方向传播。假设波长为 6 cm，请写出波的电场矢量表达式

。

7.10 对于以电场为特征的波

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,a_x\cos(\omega t - kz) + \hat{\mathbf{y}}\,a_y\cos(\omega t - kz + \delta),$$

识别偏振态，确定偏振角 (γ, χ)，绘制 E(0,t) 轨迹，并使用模块 7.3 针对以下每种情况验证您的解决方案。

(a) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 0$

(b) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = 180°$

(c) $a_x = 3$ V/m, $a_y = 3$ V/m, and $\delta = 45°$

(d) $a_x = 3$ V/m, $a_y = 4$ V/m, and $\delta = -135°$

7.11 The electric field of a uniform plane wave propagating in free space is given by

$$\widetilde{\mathbf{E}} = (\hat{\mathbf{x}} + j\hat{\mathbf{y}})30e^{-j\pi z/6} \quad \text{(V/m)}.$$

Specify the modulus and direction of the electric field intensity at the $z = 0$ plane at $t = 0$, 5, and 10 ns.

*7.12 The magnetic field of a uniform plane wave propagating in a dielectric medium with $\varepsilon_r = 36$ is given by

$$\widetilde{\mathbf{H}} = 30(\hat{\mathbf{y}} + j\hat{\mathbf{z}})e^{-j\pi x/6} \quad \text{(mA/m)}.$$

Specify the modulus and direction of the electric field intensity at the $x = 0$ plane at $t = 0$ and 5 ns.

7.13 A linearly polarized plane wave of the form $\widetilde{\mathbf{E}} = \hat{\mathbf{x}} a_x e^{-jkz}$ can be expressed as the sum of an RHC-polarized wave with magnitude a_R and an LHC-polarized wave with magnitude a_L. Prove this statement by finding expressions for a_R and a_L in terms of a_x.

*7.14 The electric field of an elliptically polarized plane wave is given by

$$\mathbf{E}(z,t) = [-\hat{\mathbf{x}}\,10\sin(\omega t - kz - 60^\circ) + \hat{\mathbf{y}}\,30\cos(\omega t - kz)] \quad \text{(V/m)}.$$

Determine:

(a) the polarization angles (γ, χ), and

(b) the direction of rotation.

7.15 Compare the polarization states of each of the following pairs of plane waves:

(a) Wave 1: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) + \hat{\mathbf{y}}2\sin(\omega t - kz)$
Wave 2: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) + \hat{\mathbf{y}}2\sin(\omega t + kz)$

(b) Wave 1: $\mathbf{E}_1 = \hat{\mathbf{x}}2\cos(\omega t - kz) - \hat{\mathbf{y}}2\sin(\omega t - kz)$
Wave 2: $\mathbf{E}_2 = \hat{\mathbf{x}}2\cos(\omega t + kz) - \hat{\mathbf{y}}2\sin(\omega t + kz)$

7.16 Plot the locus of $\mathbf{E}(0,t)$ for a plane wave with

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\sin(\omega t + kz) + \hat{\mathbf{y}}2\cos(\omega t + kz).$$

Determine the polarization state from your plot.

Sections 7-4: Propagation in a Lossy Medium

7.17 For each of the following combinations of parameters, determine if the material is a low-loss dielectric, a quasi-conductor, or a good conductor. Then calculate α, β, λ, u_p, and η_c.

*(a) Glass with $\mu_r = 1$, $\varepsilon_r = 5$, and $\sigma = 10^{-12}$ S/m at 10 GHz.

(b) Animal tissue with $\mu_r = 1$, $\varepsilon_r = 12$, and $\sigma = 0.3$ S/m at 100 MHz.

(c) Wood with $\mu_r = 1$, $\varepsilon_r = 3$, and $\sigma = 10^{-4}$ S/m at 1 kHz.

7.18 Dry soil is characterized by $\varepsilon_r = 2.5$, $\mu_r = 1$, and $\sigma = 10^{-4}$ (S/m). At each of the following frequencies, determine if dry soil may be considered a good conductor, a quasi-conductor, or a low-loss dielectric. Then calculate α, β, λ, μ_p, and η_c:

(a) 60 Hz

(b) 1 kHz

(c) 1 MHz

(d) 1 GHz

*7.19 In a medium characterized by $\varepsilon_r = 9$, $\mu_r = 1$, and $\sigma = 0.1$ S/m, determine the phase angle by which the magnetic field leads the electric field at 100 MHz.

7.20 Generate a plot for the skin depth δ_s versus frequency for seawater for the range from 1 kHz to 10 GHz (use log-log scales). The constitutive parameters of seawater are $\mu_r = 1$, $\varepsilon_r = 80$, and $\sigma = 4$ S/m.

*7.21 Ignoring reflection at the air–soil boundary, if the amplitude of a 3 GHz incident wave is 10 V/m at the surface of a wet soil medium, at what depth will it be down to 1 mV/m? Wet soil is characterized by $\mu_r = 1$, $\varepsilon_r = 9$, and $\sigma = 5 \times 10^{-4}$ S/m.

7.22 Ignoring reflection at the air–water boundary, if the amplitude of a 1 GHz incident wave in air is 20 V/m at the water surface, at what depth will it be down to 1 μV/m? Water has $\mu_r = 1$ and at 1 GHz, $\varepsilon_r = 80$, and $\sigma = 1$ S/m.

*7.23 The skin depth of a certain nonmagnetic conducting material is 3 μm at 5 GHz. Determine the phase velocity in the material.

7.24 Based on wave attenuation and reflection measurements conducted at 1 MHz, it was determined that the intrinsic impedance of a certain medium is $28.1\angle 45^\circ$ (Ω) and the skin depth is 2 m. Determine:

(a) the conductivity of the material,

(b) the wavelength in the medium, and

(c) the phase velocity.

*7.25 The electric field of a plane wave propagating in a nonmagnetic medium is given by

$$\mathbf{E} = \hat{\mathbf{z}}25e^{-30x}\cos(2\pi \times 10^9 t - 40x) \quad \text{(V/m)}.$$

Obtain the corresponding expression for $\mathbf{H}$.

7.11 在自由空间中传播的均匀平面波的电场由下式给出

$$\widetilde{\mathbf{E}} = (\hat{\mathbf{x}} + j\hat{\mathbf{y}})30e^{-j\pi z/6} \quad \text{(V/m)}.$$

指定 z = 0 平面在 t = 0、5 和 10 ns 处的电场强度的模量和方向。

*7.12 在 $\varepsilon_r = 36$ 的介电介质中传播的均匀平面波的磁场由下式给出

$$\widetilde{\mathbf{H}} = 30(\hat{\mathbf{y}} + j\hat{\mathbf{z}})e^{-j\pi x/6} \quad \text{(mA/m)}.$$

指定 x = 0 平面在 t = 0 和 5 ns 处的电场强度的模量和方向。

7.13 $\widetilde{E} = \hat{x}\, a_x e^{-jkz}$ 形式的线偏振平面波可以表示为幅度为 a_R 的 RHC 偏振波与幅度为 a_L 的 LHC 偏振波之和。通过根据 a_x 找到 a_R 和 a_L 的表达式来证明该命题。

*7.14 椭圆偏振平面波的电场由下式给出

$$\mathbf{E}(z,t) = [-\hat{\mathbf{x}}\,10\sin(\omega t - kz - 60^\circ) + \hat{\mathbf{y}}\,30\cos(\omega t - kz)] \quad \text{(V/m)}.$$

决定：

(a) 偏振角 (γ, χ)，以及

(b) 旋转方向。

7.15 比较下列各对平面波的偏振态：

(a) 第 1 波：$E_1 = \hat{x}\,2\cos(\omega t - kz) + \hat{y}2\sin(\omega t - kz)$ 第 2 波：$E_2 = \hat{x}\,2\cos(\omega t + kz) + \hat{y}2\sin(\omega t + kz)$ (b) 第 1 波：$E_1 = \hat{x}\,2\cos(\omega t - kz) - \hat{y}2\sin(\omega t - kz)$ 第 2 波：$E_2 = \hat{x}\,2\cos(\omega t + kz) - \hat{y}2\sin(\omega t + kz)$

7.16 绘制平面波 E(0,t) 的轨迹

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\sin(\omega t + kz) + \hat{\mathbf{y}}2\cos(\omega t + kz).$$

从图中确定偏振态。

第 7-4 节：有损介质中的传播

7.17 对于以下每个参数组合，确定该材料是低损耗电介质、准导体还是良导体。然后计算 α、β、λ、u_p 和 η_c。

*(a) 10 GHz 时具有 $\mu_r = 1$、$\varepsilon_r = 5$ 和 $\sigma = 10^{-12}$ S/m 的玻璃。

(b) 100 MHz 时 $\mu_r = 1$、$\varepsilon_r = 12$ 和 $\sigma = 0.3$ S/m 的动物组织。

(c) 木材，在 1 kHz 时具有 $\mu_r = 1$、$\varepsilon_r = 3$ 和 $\sigma = 10^{-4}$ S/m。

7.18 干燥土壤的特征为 $\varepsilon_r = 2.5$、$\mu_r = 1$ 和 $\sigma = 10^{-4}$ (S/m)。在以下每个频率下，确定干燥土壤是否可以被视为良导体、准导体或低损耗电介质。然后计算 α、β、λ、μ_p 和 η_c：

(a) 60 Hz (
b) 1 kHz (c
) 1 MHz (d
) 1 GHz

*7.19 在具有 $\varepsilon_r = 9$、$\mu_r = 1$ 和 $\sigma = 0.1$ S/m 特征的介质中，确定 100 MHz 时磁场超前于电场的相位角。

7.20 生成 1 kHz 至 10 GHz 范围内海水趋肤深度 δ_s 与频率的关系图（使用双对数刻度）。海水的本构参数为$\mu_r = 1$、$\varepsilon_r = 80$和$\sigma = 4$ S/m。

*7.21 忽略空气-土壤边界处的反射，如果 3 GHz 入射波的幅度在湿土壤介质表面为 10 V/m，那么在什么深度它会降至 1 mV/m？湿土的特征为 $\mu_r = 1$、$\varepsilon_r = 9$ 和 $\sigma = 5 \times 10^{-4}$ S/m。

7.22 忽略空气-水边界处的反射，如果空气中1 GHz 入射波的振幅在水面处为20 V/m，那么在什么深度处它会降至1 μ V/m？水的 $\mu_r = 1$ 在 1 GHz、$\varepsilon_r = 80$ 和 $\sigma = 1$ S/m。

*7.23 某种非磁性导电材料的集肤深度在 5 GHz 时为 3 μ m。确定材料中的相速度。

7.24 根据在 1 MHz 下进行的波衰减和反射测量，确定某种介质的固有阻抗为 $28.1\times45^\circ$ (Ω)，集肤深度为 2 m。决定：

(a) 材料的电导率，(b) 介质中的波长，以及 (c) 相速度。

*7.25 在非磁性介质中传播的平面波的电场由下式给出

$$\mathbf{E} = \hat{\mathbf{z}}25e^{-30x}\cos(2\pi \times 10^9 t - 40x) \quad \text{(V/m)}.$$

求出H对应的表达式。

7.26 The magnetic field of a plane wave propagating in a nonmagnetic medium is given by

$$\mathbf{H} = \hat{\mathbf{y}}\,60e^{-10z}\cos(2\pi \times 10^8 t - 12z) \qquad \text{(mA/m)}.$$

Obtain the corresponding expression for **E**.

7.27 At 2 GHz, the conductivity of meat is on the order of 1 (S/m). When a material is placed inside a microwave oven and the field is activated, the presence of the electromagnetic fields in the conducting material causes energy dissipation in the material in the form of heat.

(a) Develop an expression for the time-average power per mm^3 dissipated in a material of conductivity σ if the peak electric field in the material is E_0.

(b) Evaluate the result for an electric field $E_0 = 4 \times 10^4$ (V/m).

Section 7-5: Current Flow in Conductors

7.28 In a nonmagnetic, lossy, dielectric medium, a 300 MHz plane wave is characterized by the magnetic field phasor

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \qquad \text{(A/m)}.$$

Obtain time-domain expressions for the electric and magnetic field vectors.

*__7.29__ A rectangular copper block is 30 cm in height (along z). In response to a wave incident upon the block from above, a current is induced in the block in the positive x direction. Determine the ratio of the ac resistance of the block to its dc resistance at 1 kHz. The relevant properties of copper are given in Appendix B.

7.30 Repeat Problem 7.29 at 10 MHz.

7.31 The inner and outer conductors of a coaxial cable have radii of 0.5 cm and 1 cm, respectively. The conductors are made of copper with $\varepsilon_r = 1$, $\mu_r = 1$, and $\sigma = 5.8 \times 10^7$ S/m. The outer conductor is 0.5 mm thick. At 10 MHz:

(a) Are the conductors thick enough to be considered infinitely thick as far as the flow of current through them is concerned?

(b) Determine the surface resistance R_s.

(c) Determine the ac resistance per unit length of the cable.

7.32 Repeat Problem 7.31 at 1 GHz.

Section 7-6: EM Power Density

*__7.33__ The magnetic field of a plane wave traveling in air is given by $\mathbf{H} = \hat{\mathbf{x}}\,50\sin(2\pi \times 10^7 t - ky)$ (mA/m). Determine the average power density carried by the wave.

7.34 A wave traveling in a nonmagnetic medium with $\varepsilon_r = 9$ is characterized by an electric field given by

$$\mathbf{E} = [\hat{\mathbf{y}}\,3\cos(\pi \times 10^7 t + kx) - \hat{\mathbf{z}}\,2\cos(\pi \times 10^7 t + kx)] \qquad \text{(V/m)}.$$

Determine the direction of wave travel and average power density carried by the wave.

7.35 The electric-field phasor of a uniform plane wave traveling downward in water is given by

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\,5e^{-0.2z}e^{-j0.2z} \qquad \text{(V/m)},$$

where $\hat{\mathbf{z}}$ is the downward direction and $z = 0$ is the water surface. If $\sigma = 4$ S/m,

(a) Obtain an expression for the average power density.

(b) Determine the attenuation rate.

*__(c)__ Determine the depth at which the power density has been reduced by 40 dB.

7.36 The amplitudes of an elliptically polarized plane wave traveling in a lossless, nonmagnetic medium with $\varepsilon_r = 4$ are $H_{y0} = 3$ (mA/m) and $H_{z0} = 4$ (mA/m). Determine the average power flowing through an aperture in the y–z plane if its area is 20 m^2.

*__7.37__ A wave traveling in a lossless, nonmagnetic medium has an electric field amplitude of 24.56 V/m and an average power density of 2.4 W/m^2. Determine the phase velocity of the wave.

7.38 At microwave frequencies, the power density considered safe for human exposure is 1 (mW/cm^2). A radar radiates a wave with an electric field amplitude E that decays with distance as $E(R) = (3{,}000/R)$ (V/m), where R is the distance in meters. What is the radius of the unsafe region?

7.39 Consider the imaginary rectangular box shown in **Fig. P7.39**.

(a) Determine the net power flux $P(t)$ entering the box due to a plane wave in air given by

$$\mathbf{E} = \hat{\mathbf{x}}\,E_0\cos(\omega t - ky) \qquad \text{(V/m)}.$$

*__(b)__ Determine the net time-average power entering the box.

7.40 Repeat Problem 7.39 for a wave traveling in a lossy medium in which

$$\mathbf{E} = \hat{\mathbf{x}}\,100e^{-20y}\cos(2\pi \times 10^9 t - 40y) \qquad \text{(V/m)},$$

$$\mathbf{H} = -\hat{\mathbf{z}}\,0.64e^{-20y}\cos(2\pi \times 10^9 t - 40y - 36.85^\circ) \qquad \text{(A/m)}.$$

The box has dimensions $a = 1$ cm, $b = 2$ cm, and $c = 0.5$ cm.

7.26 在非磁性介质中传播的平面波的磁场由下式给出

$$\mathbf{H} = \hat{\mathbf{y}}\,60e^{-10z}\cos(2\pi \times 10^8 t - 12z) \qquad \text{(mA/m)}.$$

求出E对应的表达式。

7.27 在 2 GHz 时，肉的电导率约为 1 (S/m)。当将材料放入微波炉内并激活磁场时，导电材料中存在的电磁场会导致材料以热量的形式耗散能量。

(a) 如果材料中的峰值电场为 E_0，则建立电导率 σ 材料中每毫米 ^{3} 耗散的时间平均功率的表达式。 (b) 评估电场 $E_0 = 4 \times 10^4$ (V/m) 的结果。

第 7-5 节：导体中的电流

7.28 在非磁性、有损耗的介电介质中，300 MHz 平面波的特征在于磁场相量

$$\tilde{\mathbf{H}} = (\hat{\mathbf{x}} - j4\hat{\mathbf{z}})e^{-2y}e^{-j9y} \qquad \text{(A/m)}.$$

获得电场和磁场矢量的时域表达式。

*7.29 矩形铜块高 30 厘米（沿 z 方向）。响应于从上方入射到块上的波，在块中沿正 x 方向感应出电流。确定 1 kHz 时模块的交流电阻与其直流电阻之比。铜的相关特性在附录B中给出。

7.30 在 10 MHz 下重复问题 7.29。

7.31 同轴电缆的内导体和外导体的半径分别为 0.5 厘米和 1 厘米。导体由 $\varepsilon_r = 1$、$\mu_r = 1$ 和 $\sigma = 5.8 \times 10^7$ S/m 的铜制成。外导体厚度为 0.5 毫米。 10 MHz 时：

(a) 就流经导体的电流而言，导体的厚度是否足以被视为无限厚？ (b) 确定表面电阻 R_s。 (c) 确定电缆单位长度的交流电阻。 7.32 在 1 GHz 下重复问题 7.31。

第 7-6 节：电磁功率密度

*7.33 在空气中传播的平面波的磁场由 $H = \hat{x}\ 50\sin(2\pi \times 10^7 t - ky)$ (mA/m) 给出。确定波所携带的平均功率密度。

7.34 在具有 $\varepsilon_r = 9$ 的非磁性介质中传播的波的特征在于电场为：

$$\mathbf{E} = [\hat{\mathbf{y}}\,3\cos(\pi \times 10^7 t + kx) - \hat{\mathbf{z}}\,2\cos(\pi \times 10^7 t + kx)] \qquad \text{(V/m)}.$$

确定波传播的方向和波所携带的平均功率密度。

7.35 在水中向下传播的均匀平面波的电场相量由下式给出

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\,5e^{-0.2z}e^{-j0.2z} \qquad \text{(V/m)},$$

其中 $\hat{z}$ 是向下方向，$z = 0$ 是水面。如果$\sigma = 4$ S/m，

(a) 获得平均功率密度的表达式。
(b) 确定衰减率。 *(c) 确定功率密度降低 40 dB 的深度。

7.36 在具有$\varepsilon_r = 4$ 的无损非磁性介质中传播的椭圆偏振平面波的振幅为$H_{y0} = 3$ (mA/m) 和$H_{z0} = 4$ (mA/m)。如果 y-z 平面上的孔径面积为 20 m^2，则确定流经该孔径的平均功率。

*7.37 在无损、非磁性介质中传播的波的电场幅度为 24.56 V/m，平均功率密度为 2.4 W/m^2。确定波的相速度。

7.38 在微波频率下，人体暴露安全的功率密度为 1 (mW/cm^2)。雷达辐射电场幅度为 E 的波，该波随距离衰减为 $E(R) = (3, 000/R)$ (V/m)，其中 R 是以米为单位的距离。不安全区域的半径是多少？

7.39 考虑图 P7.39 所示的假想矩形框。

(a) 确定由于空气中的平面波而进入盒子的净功率通量 P(t)：

$$E = \hat{x}\ E_0\cos(\omega t - ky) \qquad \text{(V/m)}.$$

*(b) 确定进入框中的净时间平均功率。

7.40 对于在有损介质中传播的波，重复问题 7.39，其中

$$\mathbf{E} = \hat{\mathbf{x}}\,100e^{-20y}\cos(2\pi \times 10^9 t - 40y) \qquad \text{(V/m)},$$

$$\mathbf{H} = -\hat{\mathbf{z}}\,0.64e^{-20y}\cos(2\pi \times 10^9 t - 40y - 36.85^\circ) \qquad \text{(A/m)}.$$

盒子的尺寸为 a = 1 厘米、b = 2 厘米和 c = 0.5 厘米。